

# Pandas Tutorial 2: Reading and Exploring Real Datasets

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In this tutorial, students will learn how to read real datasets using Pandas and perform basic exploration. This is a very important step in Exploratory Data Analysis.

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## 1. Reading a CSV File

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In real projects, data is usually stored in CSV files.

CSV means Comma Separated Values. Each value in a row is separated by a comma.

To read a CSV file in Pandas:

```
import pandas as pd

df = pd.read_csv("data.csv")
```

Here:

- `pd.read_csv` is used to read CSV files.
- `"data.csv"` is the file name.
- `df` is the DataFrame where data is stored.

Make sure the file is in the same folder as your Python file. Otherwise, provide the full file path.

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## 2. Reading an Excel File

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Sometimes data is stored in Excel format.

To read an Excel file:

```
df = pd.read_excel("data.xlsx")
```

If required, install openpyxl:

```
pip install openpyxl
```

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### 3. Viewing the First and Last Rows

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After loading a dataset, the first step is to view some rows.

To view first 5 rows:

```
df.head()
```

To view last 5 rows:

```
df.tail()
```

You can also specify number of rows:

```
df.head(10)
```

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### 4. Checking Dataset Shape

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To know how many rows and columns are present:

```
df.shape
```

The output will be in this format:

(Number\_of\_rows, Number\_of\_columns)

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### 5. Getting Basic Information About Dataset

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To get summary information about columns:

```
df.info()
```

This function shows:

- Column names.
- Number of non null values.
- Data types of each column.

- Memory usage.

This is very important for understanding data structure.

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## 6. Checking Data Types

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To check data types only:

```
df.dtypes
```

This tells whether a column is:

- int
- float
- object
- bool
- datetime

Understanding data types is important for correct analysis.

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## 7. Handling Missing Values

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Missing values are very common in real datasets.

To check missing values count:

```
df.isnull().sum()
```

To drop rows with missing values:

```
df.dropna()
```

To fill missing values:

```
df.fillna(0)
```

Handling missing values is an important part of Exploratory Data Analysis.

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## 8. Basic Statistical Summary

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To get summary statistics for numeric columns:

```
df.describe()
```

This provides:

- Count
- Mean
- Standard deviation
- Minimum
- Maximum
- Percentiles

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## 9. Checking Unique Values

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To check unique values in a column:

```
df["ColumnName"].unique()
```

To count unique values:

```
df["ColumnName"].nunique()
```

This is useful for categorical data analysis.

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## 10. Value Counts

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To count frequency of values in a column:

```
df["ColumnName"].value_counts()
```

This is very useful in EDA for understanding distribution of categories.

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## 11. Conclusion of Tutorial 2

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In this tutorial, students have learned:

1. How to read CSV and Excel files.
2. How to explore dataset using head and tail.
3. How to check shape and information of data.
4. How to detect missing values.
5. How to get statistical summary.
6. How to analyze unique values and value counts.

Now students are ready to start real Exploratory Data Analysis on datasets.

In the next tutorial, we will learn about selecting rows, filtering data, and conditional operations.

This completes Pandas Tutorial 2.